

Master Thesis Presentation

Optimal Transport in Linear Independent Component Analysis

Ashutosh Jha

University of Tuebingen

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Outline

Introduction (1/3)

- Linear Independent Component Analysis (ICA) seeks to recover a set of statistically independent components from their linear mixtures.
- Unlike Principal Component Analysis (PCA), which only ensures uncorrelated components, ICA imposes the stronger condition of full statistical independence.
- This stronger criterion allows ICA to disentangle latent variables that PCA cannot fully separate.

Introduction (2/3)

- The approach builds upon intuition from the Central Limit Theorem: mixtures of independent random variables tend towards Gaussian distributions regardless of original distributions.
- Consequently, any mixture of independent **non-Gaussian** sources appears more Gaussian than the original sources themselves.
- After whitening the mixed signal to remove correlations, the problem reduces to finding a rotation on the unit sphere that maximizes non-Gaussianity.

Introduction (3/3)

- The goal is to identify directions where projected data are least Gaussian, corresponding to candidate statistically independent components.
- To formalize this as an optimization problem, we measure Gaussianity using the Wasserstein-2 (W_2) distance between the empirical distribution of projections and a Gaussian distribution.
- This distance comes from Optimal Transport theory, which quantifies the minimal “cost” of morphing one probability distribution into another.

The linear ICA model assumes that the observed signals are linear mixtures of statistically independent sources:

$$\mathbf{X} = \mathbf{AS}$$

where

$$\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{S} = \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

The sources s_i are assumed to be statistically independent, and at most one s_i can be Gaussian, since a mixture of Gaussian signals remains Gaussian and cannot be separated by ICA [?].

- Observed mixtures: $\mathbf{X} \in \mathbb{R}^d$.
- Center before further processing: $\mathbb{E}[\mathbf{X}] = \mathbf{0}$.
- Covariance formula:

$$\text{Cov}(\mathbf{X}) = \mathbb{E}[\mathbf{X}\mathbf{X}^\top] - \mathbb{E}[\mathbf{X}]\mathbb{E}[\mathbf{X}]^\top$$

- If $\mathbb{E}[\mathbf{X}] = \mathbf{0}$ then

$$\text{Cov}(\mathbf{X}) = \mathbb{E}[\mathbf{X}\mathbf{X}^\top]$$

Whitening: Decorrelating with Eigenvalue Decomposition

- Start with centered data $\mathbf{X} \in \mathbb{R}^d$.
- Compute the covariance matrix:

$$\text{Cov}(\mathbf{X}) = \mathbb{E}[\mathbf{X}\mathbf{X}^\top]$$

- Eigenvalue decomposition of covariance:

$$\text{Cov}(\mathbf{X}) = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$$

where $\mathbf{U}$: eigenvectors, $\mathbf{\Lambda}$: diagonal matrix of eigenvalues.

- Apply decorrelation transform:

$$\mathbf{W} = \mathbf{V}^\top$$

- $\mathbf{W}\mathbf{X}$ yields a diagonal covariance matrix with variance entries λ_j , but not necessarily unit variance.

Whitening: Scaling to Identity Covariance

- To "whiten" data, rescale each decorrelated dimension to have unit variance.
- Whitening transformation:

$$\mathbf{W} = \mathbf{\Lambda}^{-1/2} \mathbf{V}^T$$

Here, $\mathbf{\Lambda}^{-1/2}$ scales diagonal entries by $1/\sqrt{\lambda_j}$.

- Whitened data:

$$\tilde{\mathbf{X}} = \mathbf{W}\mathbf{X}$$

- Covariance after whitening:

$$\text{Cov}(\tilde{\mathbf{X}}) = \mathbf{W}\text{Cov}(\mathbf{X})\mathbf{W}^T = \mathbf{I}_d$$

- Eigenvalue decomposition decorrelates data; scaling by $\mathbf{\Lambda}^{-1/2}$ gives unit variance.

Whitening and Scaling

- Scaling the data so all dimensions have unit variance and are uncorrelated.
- Found (with ED/SVD) a linear $\mathbf{W}$ that gives $\tilde{\mathbf{X}} = \mathbf{W}\mathbf{X}$ s.t.

$$\text{Cov}(\tilde{\mathbf{X}}) = \mathbb{E}[\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top] = \mathbb{E}[\mathbf{W}\mathbf{X}\mathbf{X}^\top\mathbf{W}^\top] = \mathbf{W}\mathbb{E}[\mathbf{X}\mathbf{X}^\top]\mathbf{W}^\top$$

$$\text{Cov}(\tilde{\mathbf{X}}) = \mathbf{W}\text{Cov}(\mathbf{X})\mathbf{W}^\top$$

- Diagonalized covariance:

$$\text{Cov}(\tilde{\mathbf{X}}) = \mathbf{W}\text{Cov}(\mathbf{X})\mathbf{W}^\top = \mathbf{D}$$

but we choose scaling $(\mathbf{\Lambda}^{-1/2})$ such that $\mathbf{D} = \mathbf{I}_d$.

- **Why this scaling?**
 - Ensures all dimensions are comparable.
 - Reduces ICA problem to a rotation (orthogonal search, next slide).

- After whitening, seek a matrix $\mathbf{U}$ to "unmix" such that:

$$\tilde{\mathbf{Z}} = \mathbf{U}\tilde{\mathbf{X}}$$

where $\tilde{\mathbf{Z}}$ is an estimated candidate vector for independent components.

- Covariance after unmixing:

$$\text{Cov}(\tilde{\mathbf{Z}}) = \mathbf{U} \text{Cov}(\tilde{\mathbf{X}}) \mathbf{U}^T = \mathbf{U} \mathbf{I}_d \mathbf{U}^T = \mathbf{U} \mathbf{U}^T$$

- When $\mathbf{U}$ is orthogonal ($\mathbf{U}\mathbf{U}^T = \mathbf{I}_d$), recovered components are uncorrelated and ready for independence optimization.
- ICA then finds the rotation that makes elements of $\tilde{\mathbf{Z}}$ as independent as possible.

Why PCA is not sufficient?

- 1 Make sample decorrelated and whitened, i.e. $E[\mathbf{X}] = 0$, $E[\mathbf{X}\mathbf{X}^T] = \mathbf{I}$. At this point, any orientation is good enough for PCA, so it is not sufficient for ICA, when the data are non gaussian. For multivariate gaussian data decorrelation implies independence, so there is not much more for ICA to do.
- 2 For ICA, find local optima of the function $f(\mathbf{u}) = E[(\mathbf{u}^T \mathbf{X})^4]$ where $\mathbf{u} \in S_n$ (unit sphere). Gradient descent can be used to find candidate components.

- ICA aims to find a linear transform B such that $Y = BX$ has maximally independent components.
- Standard decorrelation (whitening) gives uncorrelated axes, but not full independence unless data are Gaussian.
- Information geometry relates independence and non-Gaussianity: ICA projects distributions onto product and Gaussian manifolds to quantify both phenomena [?].

Manifolds of Product and Gaussian Distributions

- $\mathcal{P}$: Product manifold – distributions where all components are independent.
- $\mathcal{G}$: Gaussian manifold – all multivariate Gaussians (zero mean, fixed covariance).
- ICA seeks a projection of the empirical distribution onto $\mathcal{P}$: the closest independent representation.
- After whitening, ICA operates on the intersection where covariance is identity and independence must be achieved via some optimization on rotated candidates.

KL Pythagorean Identity on Product Manifold

- For any random vector Y with density P_Y , mutual information is denoted by $I(Y)$:

$$I(Y) = D_{\text{KL}}(P_Y \parallel \prod_i P_{Y_i})$$

- More generally, for any product candidate $Q = \prod_i q_i$:

$$\begin{aligned} D_{\text{KL}}(P_Y \parallel Q) &= D_{\text{KL}}\left(P_Y \parallel \prod_i P_{Y_i}\right) + D_{\text{KL}}\left(\prod_i P_{Y_i} \parallel \prod_i q_i\right) \\ &= D_{\text{KL}}\left(P_Y \parallel \prod_i P_{Y_i}\right) + \sum_i D_{\text{KL}}(P_{Y_i} \parallel q_i) \\ &= I(Y) + \sum_i D_{\text{KL}}(P_{Y_i} \parallel q_i) \end{aligned}$$

- Minimizing KL to a product model first projects onto marginals; the leftover is mutual information.

KL Pythagorean Identity on Gaussian Manifold

$$D_{\text{KL}}(P_Y \| N(\Sigma)) = D_{\text{KL}}(P_Y \| N(\text{Cov } Y)) + D_{\text{KL}}(N(\text{Cov } Y) \| N(\Sigma))$$

- $N(\Sigma)$ denotes the multivariate Gaussian distribution with covariance Σ and zero mean.
- $N(\text{Cov } Y)$ is the Gaussian with the empirical mean and covariance matrix of Y .
- This formula expresses that KL divergence from P_Y to any Gaussian splits into the divergence to the best Gaussian fit (same covariance as Y) and the divergence between two Gaussian distributions (with covariances $\text{Cov } Y$ and Σ).

Combining Pythagorean Theorems

Let us combine the Pythagorean theorem for independence with the one for Gaussianity, interpreting ICA as successive geometric approximations:

- For an N -vector with complicated joint distribution P_Y , two simplifying approximations are:
 - **Gaussian:** $P_Y^G \equiv \mathcal{N}(\text{Cov } Y)$ – best Gaussian fit (project onto $\mathcal{G}$).
 - **Product:** $P_Y^P \equiv \prod_i P_{Y_i}$ – distribution with independent marginals (project onto $\mathcal{P}$).
- Applying both approximations yields $P_Y^{PG} \equiv \mathcal{N}(\text{diag Cov } Y)$, a Gaussian with independent marginals matching those of Y .
- These yield two triangles through distribution space:
 $[P_Y \rightarrow P_Y^G \rightarrow P_Y^{PG}]$ and $[P_Y \rightarrow P_Y^P \rightarrow P_Y^{PG}]$, sharing the hypotenuse $[P_Y \rightarrow P_Y^{PG}]$.

Information Geometry in ICA

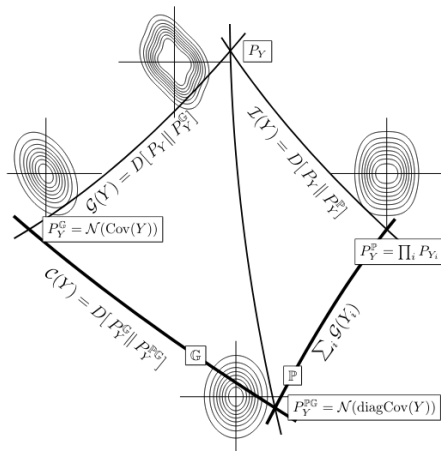


Figure 2: A probability density P_Y for a vector Y can be approximated as having independent components (approximation P_Y^P) or as being Gaussian (approximation P_Y^G) or both as P_Y^{PG} . These approximations correspond to projections onto exponential manifolds. Those densities form two ‘right triangles’, each giving rise to a Pythagorean theorem, and sharing a common hypotenuse, thus relating the ‘lengths’ of the other sides, leading to Eq. (12). The lengths of all sides have a clear and simple statistical meaning, allowing to connect independence, correlation and non Gaussianity in a single information-geometric picture.

Combining Pythagorean Theorems

The KL divergence to the product-Gaussian approximation admits two complementary decompositions:

$$D[P_Y \| P_Y^{PG}] = D[P_Y \| P_Y^P] + D[P_Y^P \| P_Y^{PG}] = D[P_Y \| P_Y^G] + D[P_Y^G \| P_Y^{PG}]$$

- **Mutual Information:** $I(Y) = D[P_Y \| P_Y^P]$
- **Non-Gaussianity:** $G(Y) = D[P_Y \| P_Y^G]$
- **Correlation Scalar:**

$$C(Y) = D[P_Y^G \| P_Y^{PG}] = D[\mathcal{N}(\text{Cov } Y) \| \mathcal{N}(\text{diag Cov } Y)]$$

Measures how far the $\text{Cov } Y$ matrix is from just its diagonal part and appears as the natural scalar measure of the correlation between entries of Y .

Non-Gaussianity, correlation and independence

- Define non-Gaussianity of Y :

$$G(Y) = D_{\text{KL}}(P_Y \parallel N(\text{Cov}(Y)))$$

- Cardoso's pythagorean theorem relates independence, correlation, and non-Gaussianity:

$$L(Y) + \sum_i G(Y_i) = C(Y) + G(Y)$$

- Where

- $C(Y) = D_{\text{KL}}(N(\text{Cov}(Y)) \parallel N(\text{diag Cov}(Y)))$ measures how correlated the Gaussian approximation of Y and decorrelated? Y is.
- $G(Y)$ is joint non-Gaussianity; $G(Y_i)$ is marginal non-Gaussianity.

Invariant KL Divergence of Gaussian Projection

If Y undergoes an invertible linear transformation, then its Gaussian projection $N(\text{Cov } Y)$ undergoes the same transformation. The Kullback-Leibler divergence to the Gaussian projection is invariant:

$$D_{\text{KL}}(P_Y \parallel N(\text{Cov } Y)) = D_{\text{KL}}(P_{A^{-1}Y} \parallel N(A^{-1} \text{Cov } Y (A^{-1})^{\top}))$$

That is, $G(Y)$ does not change under invertible linear transformations.

Whitening Reduces ICA to Maximizing Non-Gaussianity

- After whitening, $\text{Cov}(Y) = I$, so $C(Y) = 0$: Pythagorean identity reduces to

$$L(Y) = 0 - \sum_i G(Y_i) + \text{const}$$

- ICA in whitened space therefore reduces to maximizing non-Gaussianity of the marginals, subject to orthogonality constraints.
- Contrast functions (kurtosis, negentropy, Wasserstein distances) measure this non-Gaussianity and solve ICA.
- Thus, Centering and whitening constrain data to the space where independence can be achieved solely by maximizing non-Gaussianity [?].

Motivations from Central Limit Theorem

- Let $X_1, X_2, \dots, X_n$ be independent and identically distributed (i.i.d.) random variables drawn from a non-Gaussian distribution with mean μ and variance σ^2 .
- Define the sample mean as

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

- The CLT states that as $n \rightarrow \infty$, the distribution of

$$\sqrt{n} (\bar{X}_n - \mu) \xrightarrow{d} Z \sim \mathcal{N}(0, \sigma^2)$$

where the arrow $\xrightarrow{d}$ denotes convergence in distribution.

Weighted Sum Interpretation of CLT

- The sample mean can be seen as a weighted sum:

$$\bar{X}_n = \sum_{i=1}^n \frac{1}{n} X_i$$

where all weights $\frac{1}{n}$ are equal.

- If viewed as an expectation, this weighted sum is

$$\mathbb{E}[X] = \sum_{i=1}^n w_i X_i, \quad \text{where} \quad w_i = \frac{1}{n}, \quad \sum_{i=1}^n w_i = 1.$$

- **Conclusion:** Any weighted mixture of non-Gaussian sources tends to be closer to a Gaussian distribution than the original sources themselves.

Why Independent Components Cannot be Gaussian

- The fundamental assumption in Independent Component Analysis (ICA) is that the independent components must be non-Gaussian for the model to be identifiable and ICA to work.
- Gaussian components cannot be separated since any linear combination of Gaussian variables is also Gaussian. This means the mixing matrix cannot be uniquely estimated.
- For Gaussian Multivariate distributions decorrelation implies independence, so after PCA, decorrelation, there is not much more ICA can achieve.
- Motivation from the CLT: weighted sums of independent non-Gaussian variables tend toward Gaussianity, so we seek sources that are maximally non-Gaussian.[?].

Introduction to Optimal Transport

- Optimal Transport (OT) is about moving "mass" in the most efficient way, akin to how one might move sand with minimal effort.
- The earth mover distance (EMD) quantifies this cost, measuring how much "work" it takes to transform one distribution into another.
- The problem was first formulated by Gaspard Monge in 1781, conceptualizing the minimal cost of moving soil [?].

Optimal Transport for Probability Distributions

- The core idea is to measure distances between probability measures using the cost of transporting one distribution into another.
- This led to metrics such as the Wasserstein distance, which quantify meaningful differences between distributions.
- The Kantorovich relaxation generalized Monge's formulation using transport plans, allowing greater mathematical and computational tractability [?, ?, ?].

Introducing Distance W_1

- W_1 distance (Earth Mover's Distance) measures the minimal total "effort" to move one distribution into another.
- For one-dimensional distributions with CDFs F and G , it can be written as:

$$W_1(\mu, \nu) = \int_{-\infty}^{\infty} |F(x) - G(x)| dx$$

- Intuitively, W_1 sums the absolute vertical differences between the cumulative distribution functions.

Introducing Distance W_2 and Comparison

- W_2 distance uses squared differences and is given by:

$$W_2(\mu, \nu) = \left(\int_0^1 |F^{-1}(t) - G^{-1}(t)|^2 dt \right)^{1/2}$$

where F^{-1}, G^{-1} are the quantile functions (inverse CDFs).

- W_2 is more sensitive to differences in spread (variance) and central location (mean).
- When comparing distributions with $\mathcal{N}(0, 1)$, W_2 better captures deviations than W_1 , making it more reliable for Gaussianity measures.
- W_1 can be more robust to outliers but might underestimate such deviations.

Motivation: Bounding Mixture Non-Gaussianity

- **Goal:** We want to justify why maximizing W_2 recovers independent sources.
- **Claim:** The non-Gaussianity of a mixture $w \cdot Z$ is upper bounded by the weighted sum of the non-Gaussianity of its independent parts.

•

$$W_2^2(w \cdot Z, \mathcal{N}) \leq \sum_{i=1}^d w_i^2 W_2^2(Z_i, \mathcal{N})$$

- **Interpretation?:** By the Central Limit Theorem, mixtures are "more Gaussian" (have lower W_2 distance to $\mathcal{N}$) than the sources. Maximizing the LHS forces the weight vector w to align with a single source Z_i .

Proof Construction: The Common Source

To prove the bound, we construct a specific coupling using a common source of randomness.

- Let $\mathbf{N} = (N_1, \dots, N_d)$ be a vector of d independent Standard Normal variables.
- Let T_i be the Optimal Transport map such that $(T_i)_\# \mathcal{N} = Z_i$.
- **Constructing the Mixture Variable X :**

$$X = \sum_{i=1}^d w_i T_i(N_i)$$

Since $T_i(N_i) \sim Z_i$, it follows that $X \sim w \cdot Z$.

- **Constructing the Gaussian Target Y :** Using the *same* noise vector $\mathbf{N}$:

$$Y = \sum_{i=1}^d w_i N_i$$

Since $\sum w_i^2 = 1$, it follows that $Y \sim \mathcal{N}(0, 1)$.

Proof Step: The Coupling Inequality

- The pair (X, Y) creates a valid joint distribution (coupling) $\pi_{X,Y}^{\mathbf{N}}$.
- **Definition of Wasserstein-2:** The infimum of cost over *all possible* couplings.

$$W_2^2(w \cdot Z, \mathcal{N}) = \inf_{\pi} \mathbb{E}_{\pi}[|x - y|^2]$$

- **The Inequality:** Since the infimum is the cost of the *best* plan, it must be less than or equal to the cost of *our specific* plan.

$$\underbrace{W_2^2(w \cdot Z, \mathcal{N})}_{\text{Cost of Optimal Plan}} \leq \underbrace{\mathbb{E}[|X - Y|^2]}_{\text{Cost of our constructed plan}}$$

Proof Step: Expanding the Square

Let us expand the expected squared difference $\mathbb{E}[|X - Y|^2]$:

$$\begin{aligned}|X - Y|^2 &= \left| \sum_{i=1}^d w_i T_i(N_i) - \sum_{i=1}^d w_i N_i \right|^2 \\ &= \left(\sum_{i=1}^d w_i (T_i(N_i) - N_i) \right)^2\end{aligned}$$

Squaring the sum yields diagonal terms ($i = j$) and cross terms ($i \neq j$):

$$= \sum_{i=1}^d w_i^2 (T_i(N_i) - N_i)^2 + \sum_{i \neq j} w_i w_j (T_i(N_i) - N_i)(T_j(N_j) - N_j)$$

Proof Conclusion: Vanishing Cross Terms

Taking the expectation $\mathbb{E}$:

- **Cross Terms** ($i \neq j$): vanish due to independence and zero mean.

$$\mathbb{E}[\text{Cross}] \propto \mathbb{E}[T_i(N_i) - N_i] \cdot \mathbb{E}[T_j(N_j) - N_j] = 0 \cdot 0 = 0$$

- **Diagonal Terms:**

$$\mathbb{E}[|X - Y|^2] = \sum_{i=1}^d w_i^2 \mathbb{E}[|T_i(N_i) - N_i|^2]$$

- Recall that $\mathbb{E}[|T_i(N_i) - N_i|^2]$ is exactly $W_2^2(Z_i, \mathcal{N})$.

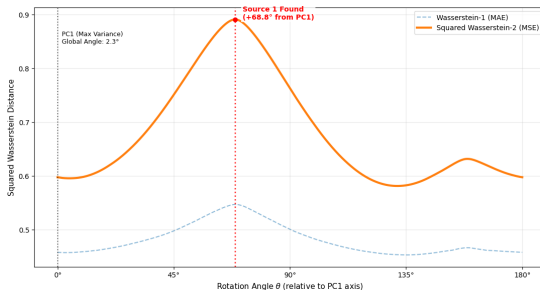
- **Result:**

$$W_2^2(w \cdot Z, \mathcal{N}) \leq \sum_{i=1}^d w_i^2 W_2^2(Z_i, \mathcal{N})$$

This convexity adjacent (triangle inequality?) property confirms that mixtures reduce the Wasserstein distance to Gaussianity.

- To select the optimal contrast function, we empirically compared the optimization landscape of W_1 (Mean Absolute Error) versus squared W_2 (Mean Squared Error).
- **Setup:**
 - Synthetic data generated from Student-t distributions.
 - We varied the degrees of freedom (ν) to control "non-Gaussianity":
 - $\nu = 2$: Heavy tails (Strong non-Gaussian signal).
 - $\nu = 10$: Approaching Normal (Weak non-Gaussian signal).
 - We plotted the distance metric over the rotation angle θ relative to the first Principal Component.

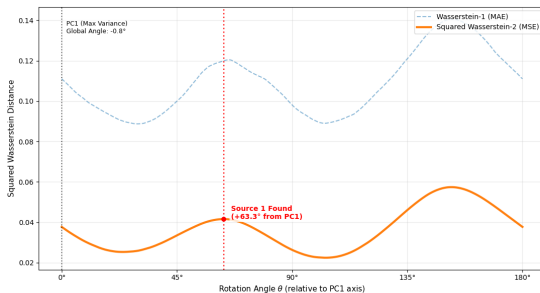
Case 1: Strong Non-Gaussianity ($\nu = 2$)



Observation:

- Both W_1 and W_2 successfully identify the source (peaks align).
- W_2 (orange) exhibits a significantly sharper peak and higher curvature around the optimum.
- This suggests W_2 provides stronger gradients for optimization when the signal is distinct.

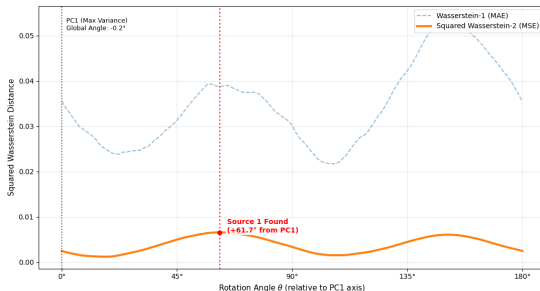
Case 2: Medium Non-Gaussianity ($\nu = 4$)



Observation:

- As tails become lighter, the signal weakens.
- W_2 maintains a smooth, convex-like profile around the optimum.
- W_1 begins to show irregularities in the landscape, though the global maximum is still recoverable.

Case 3: Weak Non-Gaussianity ($\nu = 10$)



Crucial Difference:

- The signal is now very close to Gaussian noise.
- W_1 (**blue dashed**): Becomes jagged and noisy. This "rough" landscape traps gradient-based solvers in spurious local optima.
- W_2 (**orange**): Remains smooth and unimodal.
- **Conclusion:** W_2 is slightly robust for weak signals, enabling recovery where W_1 fails.

Core Idea: Optimization Problem with W_2

- Traditional ICA formulations optimize kurtosis or negentropy contrast functions.
- Our approach formulates ICA as finding the rotation $\mathbf{u} \in S_n$ that maximizes the W_2 distance (W_2) from the standard Gaussian $\mathcal{N}(0, 1)$:

$$\hat{\mathbf{u}} = \arg \max_{\mathbf{u} \in S_n} W_2(\mathbb{P}_{\mathbf{u}^\top \mathbf{X}}, \mathcal{N}(0, 1))$$

- This perspective leverages the optimal transport metric's ability to capture deviations from Gaussianity in a geometrically meaningful way.

Core Idea: Algorithm

Algorithm 1 ICA with W2 Distance Optimization

- 1: Obtain observed mixture data $\mathbf{X}$
- 2: Perform whitening/decorrelation on $\mathbf{X}$
- 3: Initialize empty set of recovered components $\mathcal{U} = \emptyset$
- 4: **while** number of components not found **do**
- 5: Find local optimum (Gradient Descent) $\mathbf{u}^* \in S_n$ maximizing

$$W_2(\mathbb{P}_{\mathbf{u}^\top \mathbf{X}}, \mathcal{N}(0, 1))$$

- 6: Store candidate: $\mathcal{U} \leftarrow \mathcal{U} \cup \{\mathbf{u}^*\}$
- 7: Restrict search space by enforcing orthogonality:

$$\mathbf{u} \perp \text{span}(\mathcal{U})$$

- 8: **end while**
- 9: **return** $\mathcal{U}$ as estimated independent components



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